CSC4093/DSC4213: Deep Learning (2024/25)
Programming Assignment 02: Convolutional Networks and Transfer Learning

In this study, convolutional neural networks (CNNs) based on transfer learning were used to classify tea leaf images into three disease categories: *brown blight*, *algal leaf*, and *white spot*.The objective of this project is to develop and compare two deep learning models, VGG16 and ResNet50 for multi-class image classification and evaluate their performance using standard metrics.

The dataset contains 368 images across three classes: Brown Blight (113), Algal Leaf (113), and White Spot (142). It was split into 70% training, 15% validation, and 15% testing.To improve generalization and reduce overfitting, data augmentation techniques were applied to the training set, including random horizontal flipping and rotation. All images were resized to 224×224 pixels and normalized using ImageNet mean and standard deviation values.

Two pretrained CNN models were used:

VGG16

The VGG16 model pretrained on ImageNet was used as a feature extractor. The convolutional layers were frozen, and the final fully connected layer was modified to output 3 classes. Additionally, the last few convolutional layers were unfrozen to allow fine-tuning.

ResNet50

The ResNet50 model, also pretrained on ImageNet, was used with all layers frozen except the final fully connected (fc) layer. The fc layer was replaced with a new layer having 3 output neurons corresponding to the target classes.

Both models were trained using:

- Loss Function: Cross-Entropy Loss
- Optimizer: Adam
- Learning Rate: 0.001
- Batch Size: 16

A learning rate scheduler was used to improve convergence. Early stopping was applied to prevent overfitting by monitoring validation accuracy. The best model weights were saved based on validation performance.

Performance Comparison

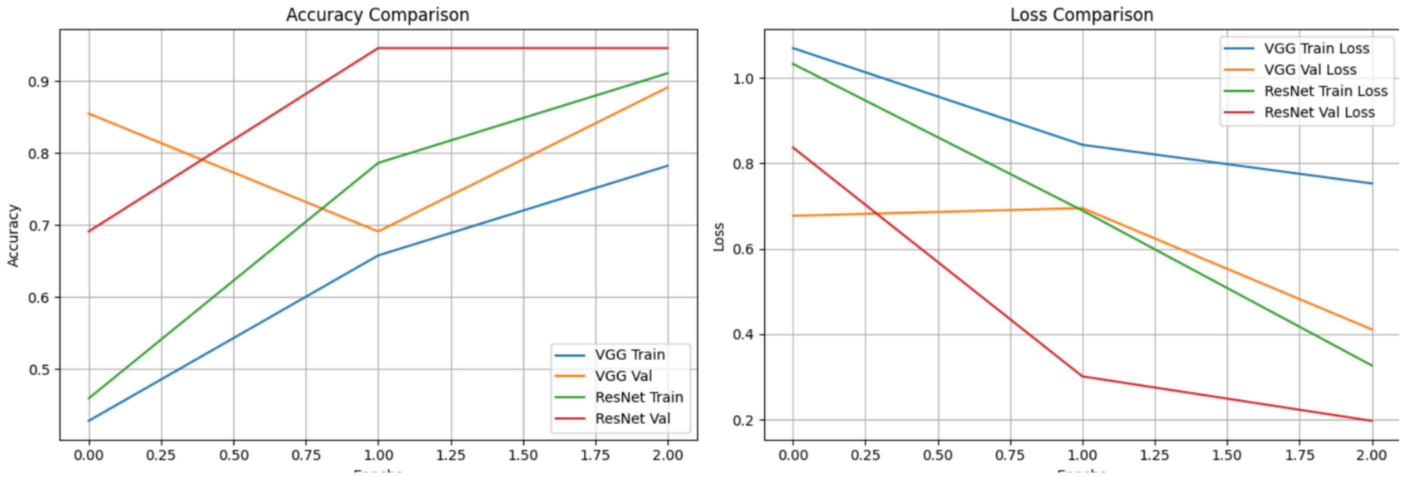
	Model	Train Accuracy	Validation Accuracy	Test Accuracy
0	VGG16	0.782101	0.890909	0.875000
1	ResNet50	0.910506	0.945455	0.892857

In addition to accuracy, classification performance was evaluated using precision, recall, and F1-score. These metrics provide a more comprehensive understanding of model performance across different classes.

	VGG16_precision	VGG16_recall	VGG16_f1-score	VGG16_support	ResNet50_precision	ResNet50_recall	ResNet50_f1-score	ResNet50_support
algal_leaf	0.894737	1.000000	0.944444	17.000	0.833333	0.882353	0.857143	17.000000
brown_blight	1.000000	0.705882	0.827586	17.000	0.933333	0.823529	0.875000	17.000000
white_spot	0.800000	0.909091	0.851064	22.000	0.913043	0.954545	0.933333	22.000000
accuracy	0.875000	0.875000	0.875000	0.875	0.892857	0.892857	0.892857	0.892857
macro avg	0.898246	0.871658	0.874365	56.000	0.893237	0.886809	0.888492	56.000000
weighted avg	0.889474	0.875000	0.872284	56.000	0.895005	0.892857	0.892496	56.000000

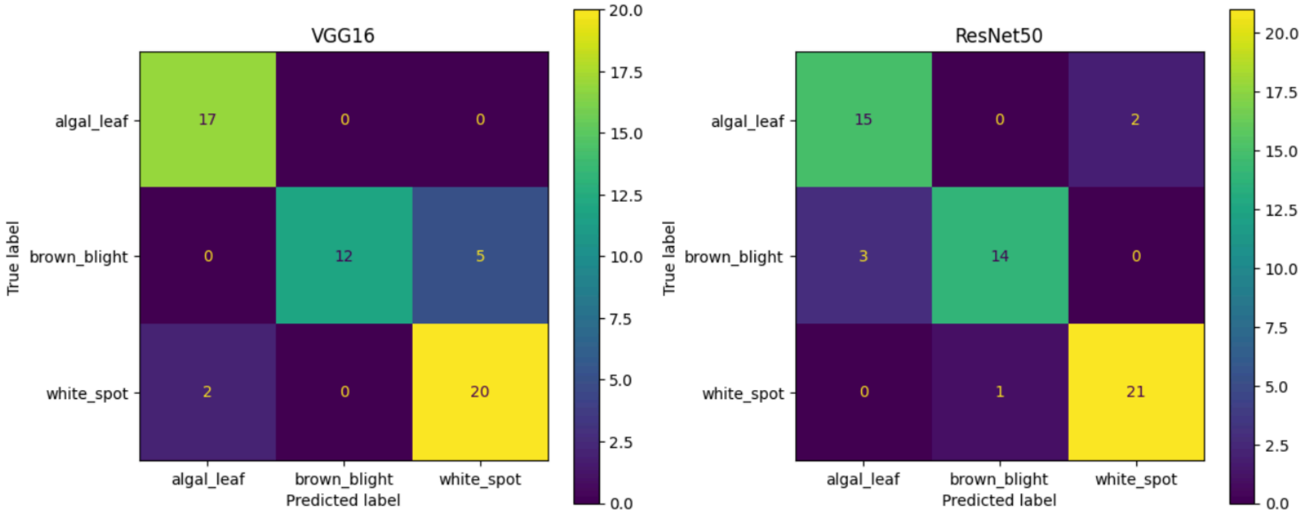
Accuracy Comparison: ResNet50's validation accuracy (red line) shoots up quickly and stabilizes near **0.95**, while VGG16's validation accuracy (orange line) actually shows a slight downward trend initially before flattening

Loss Comparison: **ResNet50** exhibits a much sharper decline in training loss (green line) and validation loss (red line), indicating faster convergence. **VGG16** shows a steadier, slower decline in training loss, but its validation loss is significantly higher than ResNet50's, ending around **0.4** compared to ResNet50's **~0.2**.



The training and validation curves indicate that both models learned effectively. ResNet50 converged faster and achieved higher validation accuracy compared to VGG16. VGG16 showed slightly higher training accuracy at later epochs, suggesting mild overfitting.

Confusion Matrices



Discussion

The results show that ResNet50 outperforms VGG16 in terms of accuracy and generalization. This improvement can be attributed to the deeper architecture of ResNet50 and the use of residual connections, which help mitigate the vanishing gradient problem and enable better feature extraction.

The dataset used in this study is relatively small, which increases the risk of overfitting. Data augmentation techniques were applied to address this limitation, but further improvements could be achieved with a larger dataset.

Additionally, early stopping played an important role in preventing over-training and improving model generalization.

Conclusion

This study demonstrates the efficacy of transfer learning for tea leaf disease classification using VGG16 and ResNet50 architectures. While both models proved capable, **ResNet50 emerged as the superior model**, achieving a peak test accuracy of **89.29%** compared to VGG16's **87.50%**. Notably, ResNet50 showed significantly stronger feature extraction during training, outperforming VGG16 by approximately **12.8%** in training accuracy.